



DATA MINING

SECP2753

ACTIVITY 1

TITLE: GROUP TASK 1

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SECTION: 02

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TASK 1

QUESTION 1

How is data mining part of the natural evolution of database technology?

- Involves extracting valuable knowledge from vast amounts of data stored in databases.
- Initially, databases focused on data storage and retrieval.
- As databases grew in size and complexity, the need for tools to analyze and extract patterns emerged.
- Data mining was developed to automatically discover patterns and relationships in large datasets.
- Fast changing and faster-paced business environment
- Faster computers, intelligent algorithms

QUESTION 2

Why is data mining important?

Wide Availability of Huge Amounts of Data: Large volumes of data are increasingly available across various industries.

Need for Turning Data into Useful Information and Knowledge:

Data alone is not useful until it's analyzed and interpreted to generate knowledge.

Applications of Data and Knowledge:

- Business management
- Production control
- Market analysis
- Engineering design
- Scientific exploration

Data Mining as a Natural Evolution:

- Data mining is a result of the development of information technology, particularly the database industry.

Stages in Data Handling:

- Data collection and database creation
- Data management (storage, retrieval, transaction processing)
- Data analysis and understanding (data warehousing, data mining)

Importance of Data Mining:

- The sheer volume of data surpasses human comprehension without powerful tools.
- Data stored in large databases often become "data tombs" or archives, rarely revisited.
- Important decisions are sometimes made based on intuition rather than on the valuable knowledge contained in the data, due to the lack of proper tools to extract insights.

QUESTION 3

What is data mining?

- Uses techniques from statistics, machine learning, and database systems.
- Involves analyzing data to extract valuable insights ,extracting or “mining” knowledge from large amounts of data.
- Other Names:knowledge mining from databases,knowledge extraction, data/pattern analysis,data archaeology,and data dredging. Knowledge Discovery in Databases (KDD)

QUESTION 4

What is the general architecture of a data mining system?

1)Database, Data Warehouse, or Other Information Repository

- Set of databases, data warehouses, spreadsheets, or other information repositories
- Data cleaning and integration techniques may be applied

2)Database or Data Warehouse Server

- Fetches relevant data based on user's data mining request

3)Knowledge Base

- Domain knowledge used to guide search or evaluate the interestingness of patterns
- Examples include additional interestingness constraints, thresholds, and metadata

4)Data Mining Engine

- Core component with modules for tasks such as characterization, association analysis, classification, evolution, and deviation analysis

5)Pattern Evaluation Module

- Uses interestingness measures to focus the search towards interesting patterns
- Interacts with the data mining modules and may access thresholds stored in the knowledge base

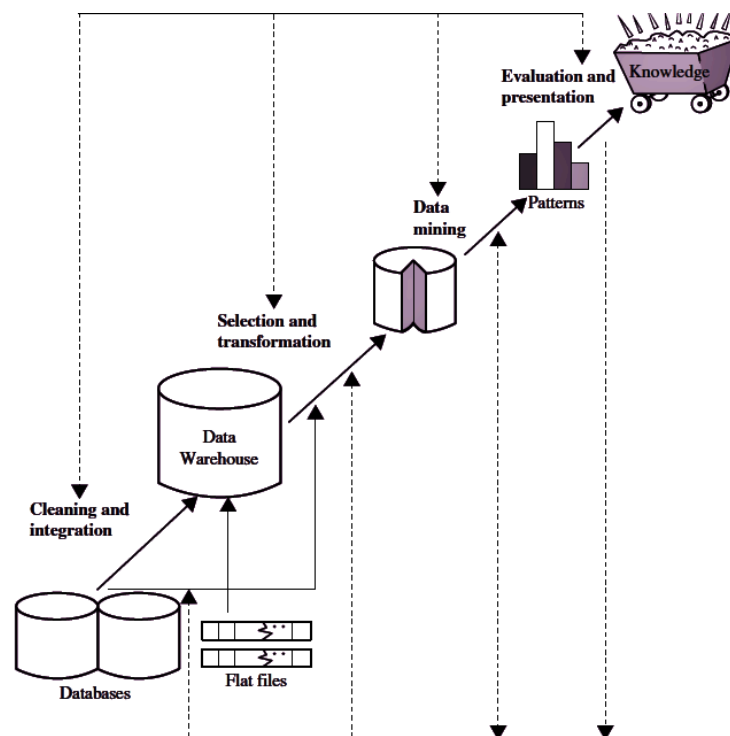
QUESTION 5

What types of data can be mined? (Structured & Unstructured data)

- Relational databases
- Data warehouses
- Transactional databases
- Advanced database systems (object-oriented and object-relational databases, spatial databases, time-series databases, text databases and multimedia databases)
- Flat files
- World-Wide Web

QUESTION 6

What are the main steps/processes involved in data mining?



- Data cleaning (to remove noise or irrelevant data)
- Data integration (where multiple data sources may be combined)
- Data selection (where data relevant to the analysis task are retrieved from the database)
- Data transformation (where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations)

- Data mining (an essential process where intelligent methods are applied in order to extract data patterns)
- Pattern evaluation (to identify the truly interesting patterns representing knowledge based on some interestingness measures)
- Knowledge presentation (where visualization and knowledge representation techniques are used to present the mined knowledge to the user)

QUESTION 7

What are the common data mining tasks/techniques and their associated algorithms?

- Classification (Nominal Label)
- Regression (Numerical Label)
- Forecasting (Numerical Label and Time-Series Data)
- Clustering (Similarity Distance Based)
- Association (Relation or Linked Based)

QUESTION 8

What issues or challenges are commonly encountered in data mining?

- Not all data mining systems on the market can perform true data mining.
- Many traditional data analysis systems are not considered a data mining system, as it struggles to handle high-dimensional or large amounts of data.
- Incomplete or inconsistent data can impact on results accuracy, causing data cleaning and preprocessing being required.
- Integrating structured and unstructured data is complex, since data comes from multiple sources.
- High-speed datasets processing causes challenges to real-time data mining.
- High implementation and maintenance costs.

QUESTION 9

What are some examples of data mining applications and tools?

1. Data Mining Applications

- Business Management (e.g. Amazon's Recommendation System)
- Production Control (e.g. Toyota Predictive Maintenance System)
- Market Analysis (e.g. Walmart's Market Basket Analysis)
- Engineering Design (e.g. Airbus Aircraft Component Optimization)
- Medical Research (e.g. IBM Watson for Cancer Diagnosis)
- Science Exploration (e.g. NASA's Exoplanet Discovery using Kepler Data)

2. Data Mining Tools

- WEKA (Open-Source Tool)
- RapidMiner (Open-Source Tool)
- Knime (Open-Source Tool)
- Tableau (Commercial Tool)
- Google Cloud AI (Commercial Tool)

QUESTION 10

What are the differences between descriptive and predictive data mining?

- **Descriptive** Data Mining : This type of data mining aims to describe and summarize existing data to identify patterns and relationships.(e.g. grouping students into different clusters based on their study hours and age to understand the different profiles of "Grade A" students)
- **Predictive** Data Mining : This type of data mining focuses on building models based on historical data to predict future outcomes or classify new, unknown data. (e.g. a decision tree of medical diagnosis is created to diagnose Strep-throat, Allergy, or Cold based on symptoms like swollen glands and fever)

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TASK 2

QUESTION 1

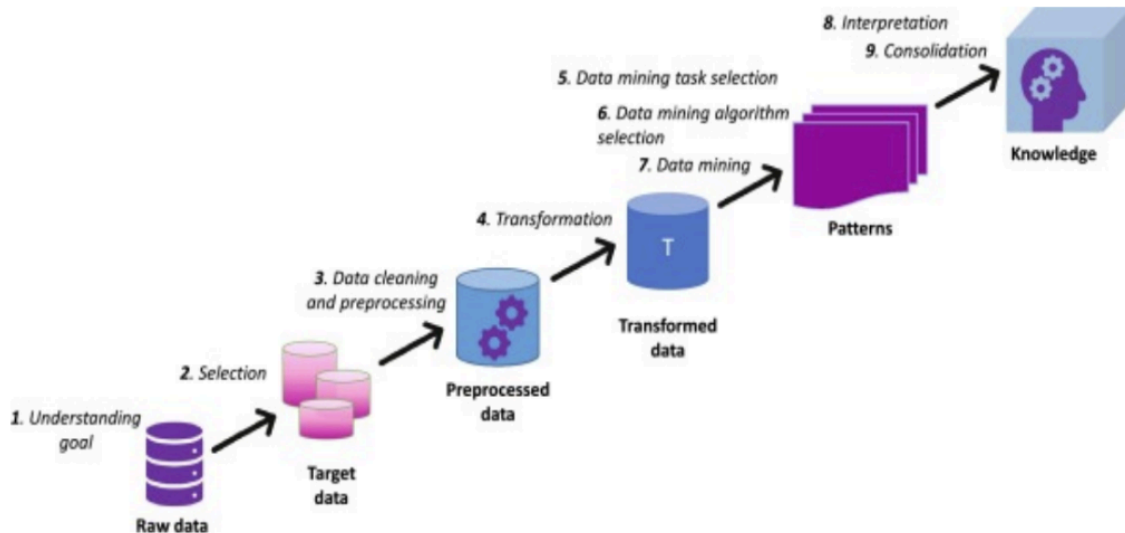
What is Knowledge Discovery in Databases (KDD)?

Knowledge discovery in databases (KDD) is a multi-step process that involves the extraction of implicit, previously unknown, and potentially useful information from data. The KDD process includes various stages, such as data cleaning, data integration, data selection, data transformation, pattern discovery, and knowledge representation. The ultimate goal of KDD is to discover new insights, patterns, and relationships that can be used to improve decision-making.

Source: Han, J., Kamber, M., & Pei, J. (2011). Data mining: concepts and techniques (3rd ed.). Morgan Kaufmann.

QUESTION 2

What are the steps in the KDD process? (Include a figure/diagram)



Step of KDD

1. Data integration:

Data integration is the process of combining data from different sources into a unified view. In this process, data is brought in from multiple sources such as files, databases, or other data repositories, and then integrated to form a cohesive and complete data set. This process involves several steps, such as identifying the data sources, cleaning and transforming the data, mapping the data to a common schema, and integrating the data into a single repository or data warehouse. The goal of data integration is to enable analysts to access and analyze data from multiple sources in a more efficient and effective way, and to gain a more complete and accurate view of the data.

Source: "Data Integration Concepts and Architecture" by Rick van der Lans. This book provides a detailed explanation of data integration and the various techniques used in data integration processes:

<https://www.amazon.com/Data-Integration-Concepts-Architecture-Management/dp/0123743190>

2. Data selection:

Data selection is the process of selecting relevant data from a larger data set for analysis. In this process, a subset of the data is chosen based on specific criteria such as data quality, relevance to the analysis task, and potential for insights and patterns. The goal of data selection is to focus the analysis on the most relevant and useful data, and to avoid analyzing irrelevant or unnecessary data.

Source: "Data Mining: Practical Machine Learning Tools and Techniques" by Ian Witten, Eibe Frank, and Mark A. Hall. This book provides a comprehensive introduction to data mining and includes a detailed explanation of data selection:

<https://www.cs.waikato.ac.nz/~ml/weka/book.html>

3. Data cleaning:

Data cleaning is the process of identifying and correcting errors and inconsistencies in data. In this process, data is examined to identify errors such as missing values, incorrect values, and inconsistencies in data values. Once errors are identified, they are corrected, which can involve replacing missing values, correcting formatting errors, and resolving inconsistencies in data values. The goal of data cleaning is to ensure that the data being analyzed is accurate, complete, and consistent.

Source: "Data Preparation for Data Mining" by Dorian Pyle. This book provides a detailed explanation of data cleaning and the various techniques used in data cleaning processes:

<https://www.amazon.com/Data-Preparation-Mining-Dorian-Pyle/dp/0123735688>

4. Data transformation:

Data transformation is the process of converting data from one format or structure to another in order to facilitate analysis. In this process, data is converted into a format that is more suitable for analysis, such as converting data from a relational format to a hierarchical format, normalizing data to eliminate redundancy, and aggregating data to a

higher level of abstraction. The goal of data transformation is to prepare the data for analysis and to make it more suitable for the specific analysis tasks.

Source: "Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, and Jian Pei. This book provides a comprehensive introduction to data mining and includes a detailed explanation of data transformation:

<https://www.amazon.com/Data-Mining-Concepts-Techniques-Management/dp/01238147>

5. Data mining:

Data mining is the process of extracting useful information and insights from large datasets. It involves the use of statistical and computational techniques to identify patterns, relationships, and trends in data that can be used to make informed decisions. The main goal of data mining is to discover knowledge that is hidden in large datasets and to use this knowledge to improve business operations, customer service, and decision-making.

Source: "Data Mining: Concepts and Techniques" by Jiawei Han, Micheline Kamber, and Jian Pei. This book provides a comprehensive introduction to data mining and includes a detailed explanation of data mining techniques:

<https://www.amazon.com/Data-Mining-Concepts-Techniques-Management/dp/01238147>

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6. Data consolidation:

Data consolidation is the process of combining data from multiple sources into a single dataset. This involves integrating data from different databases, spreadsheets, or other sources, and reconciling any differences between them. The goal of data consolidation is to create a unified and complete view of the data that can be used for analysis or other purposes.

Source: "Data Consolidation and Integration" by Michael G. Hall. This article provides a detailed explanation of data consolidation and the various techniques used in data consolidation processes:

https://www.ibm.com/support/knowledgecenter/en/SSGU8G_12.1.0/com.ibm.em.doc/concepts/c_con_data_consolidation.html

7. Pattern evaluation:

Pattern evaluation is the process of assessing the quality and usefulness of patterns that have been discovered in data mining. It involves analyzing the patterns to determine whether they are significant, actionable, and useful for achieving the goals of the analysis. The goal of pattern evaluation is to ensure that the patterns identified in data mining are valid, reliable, and relevant to the analysis task.

Source: "Pattern Evaluation" by Bing Liu. This article provides a detailed explanation of pattern evaluation and the various techniques used in pattern evaluation processes:

<https://www.cs.uic.edu/~liub/FBS/DM-chap6.pdf>

QUESTION 3

How does KDD differ from Data Mining? (KDD vs. Data Mining)

Knowledge Discovery in Databases (KDD) and **Data Mining** are closely related but distinct concepts:

KDD refers to the **entire process** of discovering useful knowledge from data. This comprehensive process includes several steps: data selection, preprocessing, transformation, data mining, and interpretation/evaluation of patterns to extract meaningful insights from large datasets.

Data Mining is a **specific step** within the KDD process. It involves applying algorithms to analyze data and extract patterns, trends, and relationships. Essentially, data mining focuses on the discovery of previously unknown information from data.

Key Difference: While KDD encompasses the whole process of knowledge discovery, data mining specifically pertains to the analysis phase where patterns are identified within the data.

QUESTION 4

What is Data and Insights?

Data: These are raw, unprocessed facts and figures without inherent meaning. Data can be in various forms such as numbers, text, or images and lack context on their own.

Insights: These are the valuable understandings or actionable knowledge derived from analyzing data. Insights provide a deeper comprehension of patterns and trends, enabling informed decision-making.

Relationship: Data serves as the foundational input which, when analyzed and interpreted, yields insights. The transformation from data to insights involves processing and contextualizing raw data to extract meaningful information.

QUESTION 5

What is the difference between data and insights?

Data	Insights
- Raw facts and figures	- Understanding you extract from data
- Unprocessed or minimally processed	- Reveals patterns, relationships and possibilities

QUESTION 6

What is CRISP-DM (Cross Industry Standard Process for Data Mining)?

CRISP-DM is a cyclical process that provides a structured approach to planning, organizing, and implementing a data mining project. The process consists of six major phases which are business

understanding, data understanding, data preparation, modeling, evaluation and deployment.

QUESTION 7

How does Information Retrieval (IR) differ from Data Mining?

Information Retrieval (IR)	Data Mining
- Search for information	- Discover information
- You know what you're looking for	- You're find new information

QUESTION 8

What is Machine Learning (ML)?

Machine learning is a branch of artificial intelligence focused on enabling computers and machines to imitate the way that humans learn which is to observe processes or phenomena and identify patterns about these observations.

QUESTION 9

What are the main types of ML?

There are two types of Machine Learning which are **supervised learning** and **unsupervised learning**. Classification (nominal label), regression (numerical label) and forecasting (numerical label and time-series data) falls under **supervised learning** whereas clustering (similarity distance based) and association (relation or linked based) falls under **unsupervised learning**.

QUESTION 10

How is DM different from ML (DM vs. ML)?

Feature	Data Mining (DM)	Machine Learning (ML)
Relationship	The adaptation of machine learning techniques to business applications.	The union of statistics and AI.

Data Scale	Often deals with large datasets.	Does not handle large amounts of data.
Goal	Aims to uncover patterns and knowledge from large datasets contributing to business strategies, knowledge bases, and scientific research.	Aims to train models that can learn from data and make predictions or decisions.
Focus	Focus on extracting or mining useful information from useful data.	Focus on learning algorithms that improve experience.

QUESTION 11

What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is built upon heuristics as opposed to statistics, attempts to apply human-thought-like processing to statistical problems.

QUESTION 12

What is the difference between ML and AI (ML vs. AI)?

Feature	Machine Learning (ML)	Artificial Intelligence (AI)
Definition	The union of statistics and AI.	Built upon heuristics as opposed to statistics, attempts to apply human-thought-like processing to statistical problems.
Goal	Enable systems to learn from data and improve over time.	Simulate human intelligence.
Example	Spotify recommends songs based on the user's past listening music choice.	Self-driving car using AI (auto-pilot).

QUESTION 13

What are some real-world applications of AI and potential careers in this field?

Example for Real-world Applications of AI

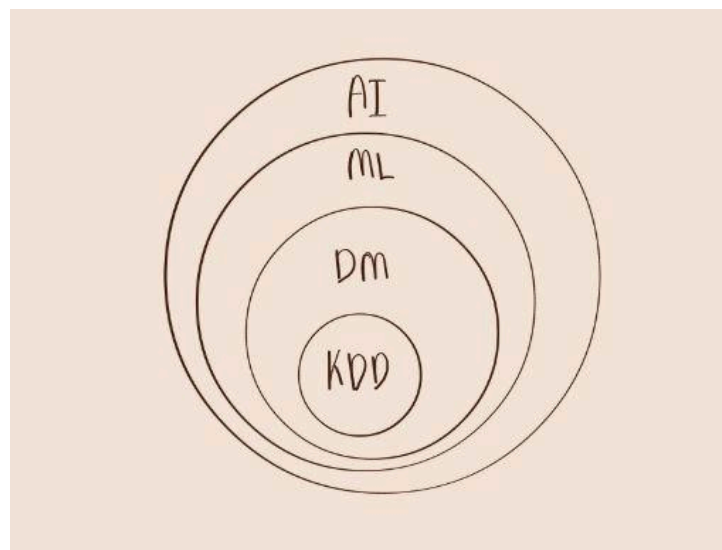
1. **Healthcare** : AI-powered imaging systems assist radiologists in detecting anomalies in X-rays and MRIs with high precision.
2. **Retail** : Recommendation systems, powered by AI suggest products to customers based on their preferences and purchasing history.
3. **Manufacturing** : AI-driven robots predict maintenance equipment downtime by forecasting failures before they occur.
4. **Agriculture** : Drones equipped with AI analyze fields to provide insights on crop health.

Potential Careers in AI field

- Data Scientist
- Robotics Engineer
- AI Ethicist

QUESTION 14

Provide a figure that illustrates the relationships among KDD, DM, ML and AI.



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